

Soft Spaces – Phase 3.2

v32.14: Degeneracy-Closed Interference Law

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Purpose

Version v32.14 tests whether the local hotspot mechanism identified in the preceding Phase 3.2 analyses survives a stricter, basis-invariant formulation. The key methodological correction is to stop decomposing the Q -space into individual eigenvectors inside degenerate eigenspaces and instead use complete degeneracy-closed projectors.

The scientific question is therefore:

Does the observed connection between hotspot prominence and constructive/destructive interference remain when each virtual Q channel is defined by a complete degenerate eigenspace?

1. Basis-invariant Q -group decomposition

Let P denote the target eigenspace and let P_G project onto a complete degeneracy-closed eigenspace G in the complementary space $Q = I - P$. For each such group, define

$$T_G = g_\eta(E - E_G) P V P_G V P, \quad (1)$$

where

$$g_\eta(x) = \frac{x}{x^2 + \eta^2}. \quad (2)$$

The effective second-order operator becomes

$$\Sigma_E = \sum_G T_G. \quad (3)$$

Because P_G is a projector onto the full degenerate eigenspace, this construction is invariant under arbitrary unitary changes of basis inside G .

2. Degeneracy-closed interference parameter

Expanding the Frobenius norm gives

$$\|\Sigma_E\|_F^2 = \sum_G \|T_G\|_F^2 + 2 \sum_{G < K} \text{Re Tr}(T_G^\dagger T_K). \quad (4)$$

This motivates the normalized interference parameter

$$\boxed{A_E^{(G)} = \frac{\|\Sigma_E\|_F^2 - \sum_G \|T_G\|_F^2}{(\sum_G \|T_G\|_F)^2}} \quad (5)$$

or equivalently

$$A_E^{(G)} = \frac{2 \sum_{G < K} \text{Re Tr}(T_G^\dagger T_K)}{(\sum_G \|T_G\|_F)^2}. \quad (6)$$

Positive values indicate net constructive operator interference; negative values indicate net cancellation.

A related basis-invariant coherence factor is

$$C_E^{(G)} = \frac{\|\Sigma_E\|_F}{\sum_G \|T_G\|_F}. \quad (7)$$

3. Frozen test design

The v32.14 test preserves the Phase 3.1/3.2 frozen model: 8 qubits, 11 Hamiltonian terms, complete degeneracy-closed eigenspaces, both perturbation families (dephasing Z/ZZ and transverse X/XX), and the 12 frozen Hamiltonian seeds 25042000–25042011.

For each seed and perturbation family, the basis-invariant quantities $A_E^{(G)}$, $C_E^{(G)}$, and the unnormalized cross term were compared with the frozen hotspot prominence. Seedwise Spearman correlations, one-sided sign tests, permutation counts, pooled correlations, and TOP/MIDDLE/BOTTOM ordering were recorded.

4. Results

Family	Metric	Median ρ	Positive seeds	Pooled ρ
Dephasing	$A_E^{(G)}$	+0.2605	10/12	+0.2651
Dephasing	$C_E^{(G)}$	+0.2500	12/12	+0.2668
Transverse	$A_E^{(G)}$	+0.2843	10/12	+0.2915
Transverse	$C_E^{(G)}$	+0.5194	12/12	+0.4080

For $A_E^{(G)}$, the one-sided sign-test value is $p = 0.01929$ in both perturbation families. The stronger coherence factor $C_E^{(G)}$ is positive in all 12 seeds in both families, with sign-test $p = 0.00024$.

The normalized TOP/MIDDLE/BOTTOM medians for $A_E^{(G)}$ were

$$\text{dephasing: } +0.529, +0.165, -0.364, \quad (8)$$

$$\text{transverse: } +0.704, +0.735, -0.385. \quad (9)$$

Strict TOP>MIDDLE>BOTTOM ordering occurred in 7/12 dephasing seeds and 3/12 transverse seeds. The mechanism is therefore statistically robust but not deterministic pair by pair.

5. Interpretation

The central result is that the interference signal survives the degeneracy-closure correction. The mechanism is therefore not an artifact of choosing a particular eigenvector basis inside a degenerate subspace.

The preferred local mechanistic picture is now

$$\Sigma_E = \sum_G T_G, \quad (10)$$

with hotspot strength increasing when the virtual $P \rightarrow Q_G \rightarrow P$ channels combine more coherently and with less cancellation.

A compact working statement is

$$\boxed{\Pi(E) \text{ tends to increase as } A_E^{(G)} \text{ and especially } C_E^{(G)} \text{ increase.}} \quad (11)$$

This is not yet a closed deterministic prominence formula. It is a basis-invariant, cross-seed statistical mechanism supported by both perturbation families.

6. Phase 3.2 status after v32.14

The Phase 3.2 evidence now has two layers:

1. **Global layer:** the $\pm E$ organization is explained by the algebraic symmetry

$$\Gamma H \Gamma^\dagger = -H, \quad (12)$$

with the corresponding projector and effective-operator covariance.

2. **Local layer:** hotspot strength is associated with constructive interference among complete degeneracy-closed virtual Q channels, captured by $A_E^{(G)}$ and $C_E^{(G)}$.

The remaining open problem is to derive, rather than only observe statistically, how this basis-invariant interference quantity enters the precise hotspot prominence amplitude.

Checkpoint conclusion

v32.14 confirms that the local interference mechanism survives a basis-invariant, degeneracy-closed reformulation. This removes an important methodological ambiguity and substantially strengthens the Phase 3.2 hotspot-mechanism result.